where $\psi$ is either the identity function or a lightweight two-layer MLP. The identity choice keeps the attention mechanism strictly linear with respect to the enriched features, while the MLP introduces an additional non-linearity. Both variants are evaluated in the ablation study. The resulting features $\{u_i\}$ are concatenated and passed to a scoring MLP that outputs unnormalized relevance logits:

$$s(x) = \mathrm{MLP}_{\text{score}}([\, u_1, u_2, \ldots, u_L \,]) \in \mathbb{R}^L.$$

These logits are normalized by a temperature-scaled softmax:

$$\alpha_i(x) = \frac{\exp(s_i(x)/\tau)}{\sum_{j=1}^{L} \exp(s_j(x)/\tau)},$$

where the temperature $\tau > 0$ controls the sharpness of the distribution (lower values encourage sparser, more decisive attention). Finally, the aggregated embedding is obtained as:

$$h_{\text{agg}} = \sum_{i=1}^{L} \alpha_i(x)\, g_i(h_i),$$

and passed to the output head for prediction.

The coefficients $\alpha_i(x)$ serve as explicit, per-input relevance scores over layers, providing a direct measure of how the model distributes representational importance across depth. Because these coefficients are learned jointly with the model, interpretability emerges as an *intrinsic* property of the architecture rather than an external post hoc explanation such as LIME [Ribeiro et al., 2016], SHAP [Lundberg and Lee, 2017], or Grad-CAM [Selvaraju et al., 2017].

**Training and integration.** LAYA is trained jointly with the backbone using the same objective (e.g., cross-entropy), without requiring auxiliary losses or multi-stage training procedures. It introduces minimal overhead: the projection adapters require $O(\sum_{i=1}^{L} d_i d^*)$ parameters, and the scoring network adds $O(Ld^{*2})$ parameters in the worst case, which remains modest for the values of $L$ and $d^*$ considered in the experiments. Since LAYA operates entirely at the output stage, the backbone requires no modification, and the module can be used as a drop-in replacement for the standard output head.

# 4 Experiments

LAYA was evaluated as a drop-in output head across both vision and language tasks, using three standard benchmarks and heterogeneous backbone architectures. The experimental setup was designed to (*i*) isolate the contribution of *depth-aware* aggregation at the output stage, (*ii*) measure performance differences against competitive baselines, and (*iii*) examine the interpretability signals exposed by the learned layer-attention weights. In addition, the interpretability analysis was extended to an artwork classification dataset, where the hierarchical nature of visual features is particularly pronounced; this setting allows assessing whether LAYA assigns greater importance to mid-level layers, which capture brushstroke- and texture-level information.

## 4.1 Experimental Setup

**Datasets and backbones.** The empirical evaluation spanned three popular and widely used benchmarks—Fashion-MNIST, CIFAR-10, and IMDB—covering low- to medium-complexity visual tasks and text classification. Although these datasets are relatively lightweight compared to large-scale industrial benchmarks, they are particularly well-suited for the scope of this work. LAYA operates strictly at the *output stage*, and its contribution must therefore be assessed *independently* of confounding factors such as large-scale pretraining, data augmentation pipelines, or high-capacity architectures. Using controlled, well-understood datasets enables isolating the effect of depth-aware aggregation while keeping the backbone fixed across methods. Moreover, these datasets span different modalities and representation hierarchies, providing a sufficiently diverse testbed for validating LAYA's generality without conflating improvements with domain-specific heuristics or model scaling. An additional interpretability-focused experiment on an artwork classification dataset further illustrates LAYA's behavior in domains where earlier visual features (e.g., texture, brushstroke) play a critical role, highlighting its ability to shift attention across depth levels based on the task.

Fashion-MNIST [Xiao et al., 2017] contains 60,000 training and 10,000 test grayscale images across 10 clothing categories at $28\times28$ resolution. For this dataset, a three-layer MLP with hidden sizes $[512, 256, 128]$ was used, with each layer followed by LayerNorm [Ba et al., 2016] and GELU activation [Hendrycks, 2016]. All intermediate activations were made available to LAYA.

CIFAR-10 [Krizhevsky and Hinton, 2009] comprises 50,000 training and 10,000 test color images in 10 object classes at 32×32 resolution. A lightweight CNN backbone was adopted, consisting of three sequential depthwise–pointwise convolutional stages [Howard et al., 2017], each followed by LayerNorm and GELU, with spatial downsampling in the latter stages. Global average pooled representations from each stage served as depth-specific features for LAYA.

For IMDB sentiment analysis [Maas et al., 2011], which includes 25,000 labeled movie reviews (positive/negative), text was processed using a trainable embedding layer followed by stacked bidirectional LSTMs [Hochreiter and Schmidhuber, 1997]. The final hidden state of each BiLSTM layer was used as its corresponding layer representation.

To probe layer-wise interpretability in settings where perceptual hierarchies are especially prominent, an artwork classification dataset was also considered. Specifically, the *Best Artworks of All Time* collection[1] was used, which contains digitized paintings by a diverse set of artists, each with metadata including artistic period, biography, and stylistic descriptors. Only the images were employed, and the data were reorganized into a multi-class classification task aimed at capturing broad stylistic patterns rather than author identification. For this experiment, a pretrained ViT-Base/16 backbone [Dosovitskiy, 2020] (initialized from ImageNet-21k weights) was adopted and hidden states from all transformer encoder blocks were exposed to LAYA. This configuration provides a suitable testbed for assessing whether the aggregation mechanism naturally reallocates attention toward shallower or mid-level layers when fine-grained perceptual cues—such as brushstroke texture or local contrast patterns—play a decisive role. Moreover, it demonstrates the broader applicability of LAYA to more complex, pretrained architectures, indicating that the proposed output-level aggregation can be seamlessly integrated across a wide range of tasks and model families.

**Baselines.** To contextualize the contribution of depth-aware aggregation, three standard output heads were considered, differing only in how they combine the layer-wise representations $\{h_i\}_{i=1}^{L}$, while all backbones and training settings were kept identical:

1. LASTLAYER. Only the deepest representation is used for classification:

$$\hat{y} = \phi(W h_L + b).$$

   This corresponds to the conventional output design adopted in most deep architectures.

2. CONCAT. Each layer representation is first mapped to a shared dimensionality via a learnable adapter $g_i(\cdot)$; the resulting vectors are concatenated and passed through a small MLP prior to classification:

$$h_{\text{agg}} = \text{MLP}_{\text{post}}\big([\, g_1(h_1),\, \ldots,\, g_L(h_L)\,]\big), \qquad \hat{y} = \phi(W h_{\text{agg}} + b).$$

   This produces a uniform, input-independent fusion of multi-depth features.

3. SCALARMIX [Peters et al., 2018]. A set of learnable scalar weights determines each layer's overall contribution on average across the dataset:

$$h_{\text{agg}} = \sum_{i=1}^{L} \alpha_i\, g_i(h_i), \qquad \alpha_i = \frac{e^{s_i}}{\sum_{j=1}^{L} e^{s_j}}, \quad s_i \in \mathbb{R}.$$

   SCALARMIX is a widely adopted baseline in NLP, particularly in contextual representations such as ELMo, where it effectively balances information across depth. It preserves layer identities and performs multi-layer fusion, making it the closest baseline to LAYA conceptually; however, the weighting is *global* and does not adapt to individual inputs.

**Training details.** All models were implemented in TensorFlow/Keras and trained with mini-batch optimization. The optimizer was Adam [Kingma and Ba, 2015] with learning rate in the range $3\times10^{-4}$–$10^{-3}$, cross-entropy as the loss, accuracy as the primary metric, and a fixed 10% validation split. Early stopping on validation accuracy with patience 5 was applied, restoring the best checkpoint, and training was capped at 50 epochs. Experiments were repeated over five fixed seeds, and results are reported as mean ± standard deviation across runs. Inputs were normalized to $[0, 1]$ and flattened for Fashion-MNIST, channel-wise normalized for CIFAR-10, and tokenized to a 20k vocabulary for IMDB.

The LAYA head was tuned by grid search independently for each dataset. For Fashion-MNIST, the search spanned $d^* \in \{64, 96, 128\}$, $\tau \in \{0.5, 1.0, 1.5\}$, $\psi \in \{\texttt{identity}, \texttt{mlp}\}$, and MLP scorer width in $\{d^*, 2d^*\}$. For CIFAR-10, the same settings were used except for $d^* \in \{128, 192, 256\}$. For IMDB, the search considered $d^* \in \{64, 96\}$ and $\tau \in \{0.7, 1.0, 1.3\}$ with the same choices for $\psi$ and scorer width. Each configuration was evaluated across three seeds, and the best configuration was selected based on the highest mean validation accuracy. The CONCAT and SCALARMIX baselines underwent parallel ablations over $d^*$ to ensure comparability.

[1] `https://www.kaggle.com/datasets/ikarus777/best-artworks-of-all-time`

Following hyperparameter tuning, all heads (LASTLAYER, CONCAT, SCALARMIX, and LAYA) were trained under identical settings and evaluated over the full five-seed protocol. Alongside accuracy, macro-F1 and two-sided $95\%$ confidence intervals were computed using $t$-statistics. To support interpretability analysis, attention weights $\alpha$ were also extracted.

All models shared the same backbone architecture, allowing the effect of the output mechanism to be isolated. As with any parametric model, the adequate capacity of LAYA depended on the intrinsic complexity of each dataset; tuning, therefore, corresponded to evaluating the method in the regime in which it was intended to operate, rather than to additional optimization. Baselines were sized according to the same principle.

For the artwork classification experiment, as mentioned earlier, a pretrained ViT-Base/16 backbone [Dosovitskiy, 2020] (initialized from ImageNet-21k weights) was coupled with a LAYA head operating on the CLS token of all encoder blocks. The backbone parameters were kept frozen, so that only the LAYA aggregation module and the final classifier were optimized, allowing the analysis to focus on depth-aware aggregation rather than on representation learning. The *Best Artworks of All Time* images were split into training, validation, and test sets, with an 80/10/10 stratified split across style labels. All images were resized to $224\times224$ and normalized to $[0, 1]$. Optimization was carried out with the Adam optimizer, a learning rate of $3\times10^{-5}$, and early stopping on validation accuracy with patience 3, for a maximum of 15 epochs. No additional data augmentation was applied to preserve the stylistic characteristics of each artwork. This experiment was conducted with a single ViT+LAYA configuration and was used solely to study layer-wise attention profiles in a setting where fine-grained visual style plays a central role, rather than to benchmark alternative heads.

### 4.2 Quantitative Results

**Benchmarking.** The quantitative comparison across the three benchmarks is reported in Table 1. In all cases, the backbone architecture was held fixed and only the output aggregation head was varied. Across Fashion-MNIST, CIFAR-10, and IMDB, LAYA consistently achieved the highest mean accuracy and macro-F1 scores. On Fashion-MNIST, LAYA reached $0.8834 \pm 0.0046$ accuracy, slightly exceeding LASTLAYER ($0.8829 \pm 0.0063$) and outperforming both SCALARMIX and CONCAT. On CIFAR-10, LAYA obtained $0.7212 \pm 0.0060$ accuracy, marginally higher than CONCAT ($0.7155 \pm 0.0096$), SCALARMIX ($0.7144 \pm 0.0171$), and LASTLAYER ($0.7112 \pm 0.0236$). A similar pattern was observed on IMDB, where LAYA reached $0.8707 \pm 0.0035$, slightly but consistently surpassing all baselines.

Notably, the 95% confidence intervals indicate that the improvements were stable across seeds. Since no data augmentation or task-specific optimization was applied and all methods were trained under identical protocols, these differences can be attributed directly to the aggregation mechanism rather than to variations in model capacity, preprocessing, or optimization schedules. The gains, therefore, reflect the advantage of allowing the output head to dynamically reweight intermediate representations instead of committing solely to the deepest one.

Two systematic trends emerged across datasets. First, although the improvements were generally modest, they tended to be slightly more noticeable on CIFAR-10, a dataset that benefits from multi-level abstraction but remains challenging for small custom models without augmentation. Second, LAYA exhibited consistently lower variance across random seeds, indicating improved stability in the resulting solutions. Both observations support the hypothesis that depth-aware aggregation provides a robust, adaptable representation at the output stage, even though the absolute differences in accuracy are small.

**Ablation study.** Table 2 reports the best-performing LAYA configurations identified for each dataset during the grid search tuning procedure described above. The projection dimension $d^*$ and the attention temperature $\tau$ varied moderately across datasets, whereas the choice of the transformation $\psi(\cdot)$ affected only IMDB, where a small MLP yielded a measurable improvement. The scorer width followed the scaling of $d^*$ but did not require large values. These results indicate that LAYA is not overly sensitive to architectural hyperparameters and achieves its gains through depth-aware aggregation rather than increased model capacity.

### 4.3 Interpretability

Beyond aggregate metrics, the analysis also considered the interpretability signals encoded in LAYA's layer-wise attention weights. Performance, in fact, is not the primary benefit of LAYA: its layer-wise attention scores provide an intrinsic explanation of how different abstraction levels support each prediction, a capability absent from standard output heads. For every input sample $x$, LAYA outputs both a class-probability vector and a set of attention coefficients $\alpha(x) \in \mathbb{R}^L$, which quantify the relative contribution of each layer to the final prediction.

**Global layer-wise attention profiles.** To analyze how LAYA distributes representational importance across depth, global statistics of the attention coefficients $\alpha_i(x)$ were computed for each hidden layer $i$ on the full test sets of